

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 6: SOUND

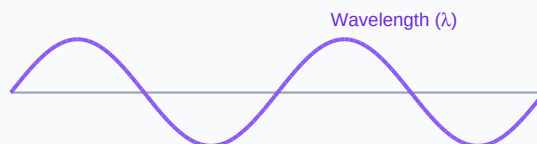
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Sound

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: PRODUCTION AND PROPAGATION OF SOUND



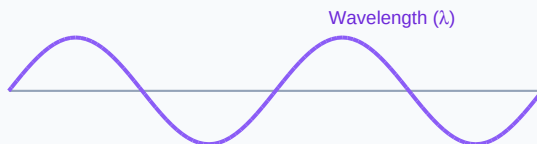
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Sound Wave	A longitudinal mechanical wave that propagates through a elastic medium via compression (high pressure) and rarefaction (low pressure) regions.
Frequency	The number of compressions or rarefactions that cross a point per unit time, measured in Hertz (Hz).
Wavelength	The distance between two consecutive compressions or two consecutive rarefactions, denoted by λ (lambda).
Echo	The reflection of sound that reaches the listener's ear at least 0.1 seconds after the direct sound, creating a distinct repetition.
Ultrasonic Waves	Sound waves with frequencies greater than 20 kHz, which cannot be heard by human ears but are used in scanning.

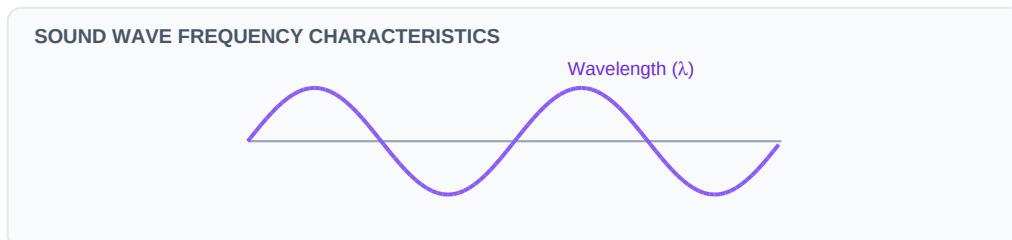
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

GLOSSARY: LONGITUDINAL ACOUSTIC WAVE CYCLES



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Wave Velocity: $v = \text{frequency (f)} \times \text{wavelength } (\lambda)$
- Time Period (T) = $1 / \text{frequency (f)}$
- Minimum distance for Echo: $d = v \times t / 2 = 344 \times 0.1 / 2 = 17.2 \text{ meters (at } 22^\circ\text{C)}$
- Sonar distance calculation: $2 \times d = v \times t$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A sound wave has a frequency of 2 kHz and a wavelength of 35 cm. How long will it take to travel 1.5 km?

Step-by-step Solution:

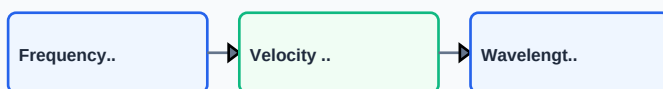
Given: $f = 2000 \text{ Hz}$, $\lambda = 0.35 \text{ m}$. Velocity $v = f * \lambda = 2000 * 0.35 = 700 \text{ m/s}$. Time to travel distance (1500 m): $t = \text{distance} / \text{velocity} = 1500 / 700 = 2.14 \text{ seconds}$.

Example 2: A ship sends out ultrasound that returns from the seabed after 4 seconds. If the speed of sound in seawater is 1500 m/s, calculate the depth.

Step-by-step Solution:

Given: $t = 4 \text{ s}$, $v = 1500 \text{ m/s}$. $2 * d = v * t \Rightarrow 2 * d = 1500 * 4 \Rightarrow 2d = 6000 \Rightarrow d = 3000 \text{ meters (3 km)}$.

WORKED: SOUND WAVE SPEED AND WAVELENGTH



HOTS (Higher Order Thinking Skills) Challenge

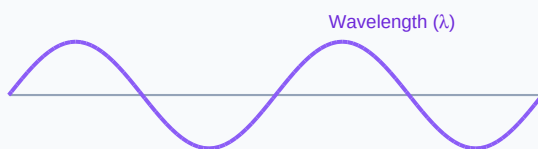
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why is sound heard more clearly and over longer distances during a humid night than on a dry hot day?

Analytical Solution Strategy:

Humid air contains water vapor, which reduces the density of air compared to dry air. Since the velocity of sound is inversely proportional to the square root of the density of the medium (Laplace correction), sound travels faster in humid air, scattering less and traveling further.

HOTS: ULTRASOUND WAVES ECHOLOCATION REFLECTIONS



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Explain the propagation of sound waves as compressions and rarefactions.

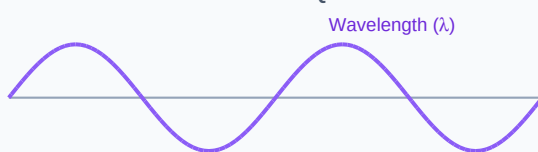
Q2 (2-Mark): Explain the three characteristics of sound: pitch, loudness, and quality/timbre.

Q3 (2-Mark): What is reverberation? How can it be reduced in auditoriums?

Q4 (5-Mark): Explain the working of SONAR with a neat diagram.

Q5 (5-Mark): Write a brief note on the audible range of human hearing, infrasound, and ultrasound.

PRACTICE: GRAPHING SOUND AMPLITUDE AND FREQUENCY



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Ultrasonography

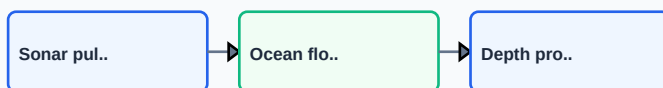
Medical scanners use high-frequency ultrasound pulses to create images of internal organs, checking fetal growth or tumor positions.

Application Case: Seabed Mapping

Submarines and research vessels use SONAR to map depth contours and locate shipwrecks.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: MARINE SONAR DEPTH NAVIGATION



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: ACOUSTIC WAVE PROPERTIES SUMMARY

